[1] you can find more on this topic in appendix C.

[2] Specifically, at least one method to add new element to the DS, and one method either to retrieve a specific element or to query the DS.

[3] In modern architectures/languages, it is possible that an array element corresponds to a *word* of memory rather than a byte, but for the sake of simplicity let's just assume an array of char is stored as an array of bytes.

[4] In principle, it doesn't need to be anything computer science: for instance, you could describe as a system a stack of files to be examined, or – a common example in computer science classes – a pile of dishes to be washed.

[5] Application programming interface

[6] A greedy algorithm is a strategy to solve problems that finds the optimal solution by making the locally optimal choice at each step. It can only find the best solution on a small subclass of problems, but it can also be used as a heuristic to find approximate (sub-optimal) solutions.

[7] NP-complete problems are a set of problems for which any given solution can be verified quickly (in polynomial time), but there is no known efficient way to locate a solution in the first place. NP-complete problems, by definition, can't be solved in polynomial time on a classical deterministic machine (for instance the RAM model we'll define in next chapter).

[8] Dynamic programming is a strategy for solving complex problems with certain characteristics: a recursive structure of subproblems, with the result of each subproblem needed multiple times in the computation of the final solution. The final solution is computed by breaking the problem down into a collection of simpler subproblems, solving each of those subproblems just once, and storing their solutions.